

Homework #2 - MATH 261A

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Problem 1

We assume that $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$ and are I.I.D.

Part (a)

We know that $\bar{X}_n$ follows a Gaussian distribution, but we must determine the mean and variance parameters.

We do this by a straightforward expectation and variance calculation:

$$E(\bar{X}_n) = E\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{1}{n} \sum_{i=1}^n E(X_i)$$

Since each X_i follows a $N(\mu, \sigma^2)$, its mean is simply μ . Therefore, we can rewrite the last sum in the above calculation as:

$$\frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \frac{\mu \cdot n}{n} = \mu$$

Therefore, we now know that the expectation of $\bar{X}_n$ is μ .

Now we calculate its variance in a similar manner:

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i)$$

We note that the third equality is valid because since the X_i are *independent* as well as identically distributed. This means that any and all covariance terms are precisely zero. Continuing the calculation, we arrive at:

$$\frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2 \cdot n}{n^2} = \frac{\sigma^2}{n}$$

With this result, we can finally conclude that $\bar{X}_n \sim N(\mu, \frac{\sigma^2}{n})$.

Part (b)

The general structure of a confidence interval is following:

$$(\text{point estimate} \pm \text{critical value} \cdot \text{standard deviation/error})$$

We are specifically deriving a 95% confidence interval for the sampling distribution of $\bar{X}_n$ which we now know is $N(\mu, \frac{\sigma^2}{n})$.

The first thing we recognize is that μ is the parameter, whose location we are trying to ascertain with the estimate $\bar{X}_n$, for a given sample of size n . Thus, the point estimate for μ is $\bar{X}_n$.

Secondly, we must determine the critical value. This is dependent on the sampling distribution which is $N(\mu, \frac{\sigma^2}{n})$. Since we know not only our distribution is Normal but the population variance too, our critical value is a z-score with form:

$$z_{\text{crit}} = \frac{\bar{x}_n - \mu}{\sigma/\sqrt{n}}$$

This expression follows a standard Normal distribution $N(0, 1)$.

The standard error is easy to find as it is simply $\frac{\sigma}{\sqrt{n}}$.

Starting at the requirement that for a 95% confidence interval, the following must be true:

$$P(-1.96 < \frac{\bar{x}_n - \mu}{\sigma/\sqrt{n}} < 1.96) = 0.95$$

To arrive at our desired confidence interval, we make take the inequality inside the probability operation and solve for μ :

$$\begin{aligned}
-1.96 &< \frac{\bar{x}_n - \mu}{\sigma/\sqrt{n}} < 1.96 \\
-1.96 \cdot \frac{\sigma}{\sqrt{n}} &< \bar{x}_n - \mu < 1.96 \cdot \frac{\sigma}{\sqrt{n}} \\
-\bar{x}_n - 1.96 \cdot \frac{\sigma}{\sqrt{n}} &< -\mu < -\bar{x}_n + 1.96 \cdot \frac{\sigma}{\sqrt{n}} \\
\bar{x}_n - 1.96 \cdot \frac{\sigma}{\sqrt{n}} &< \mu < \bar{x}_n + 1.96 \cdot \frac{\sigma}{\sqrt{n}}
\end{aligned}$$

Therefore:

$$P\left(\bar{X}_n - 1.96 \cdot \frac{\sigma}{\sqrt{n}} < \mu < \bar{x}_n + 1.96 \cdot \frac{\sigma}{\sqrt{n}}\right) = 0.95$$

Part (c)

We now assume that there is another I.I.D. sample $X_{n+1}, \dots, X_{2n}$ of size n , that is also independent of the first sample, and is from another Normal distribution $N(\mu, \tau^2)$. (We also assume that τ^2 does **NOT** refer to the precision, but instead the variance of the second sample.)

Immediately, we can conclude that the sampling distribution of $\bar{X}_{2n}$ is yet another Normal distribution. Now, we must find the expectation and variance parameters, as before.

The expectation is straightforward after the following realization: when we calculated the expectation of the sample average, the sample size appeared in the numerator and the denominator of the fraction, which meant that it was cancelled out. This means that the expectation of this new statistic $\bar{X}_{2n}$ is also μ . So will any other statistic that involved taking the average of k number of variables.

The variance is best found by calculating it directly:

$$\text{Var}(\bar{X}_{2n}) = \text{Var}\left(\frac{\sum_{i=1}^{2n} X_i}{2n}\right) = \frac{1}{4n^2} \sum_{i=1}^{2n} \text{Var}(X_i) = \frac{1}{4n^2} \sum_{i=1}^{2n} \sigma^2 = \frac{\sigma^2 \cdot 2n}{4n^2} = \frac{\sigma^2}{2n}$$

Notice that the expression found for $\text{Var}(\bar{X}_{2n})$ is the exact same as $\text{Var}(\bar{X}_n)$ except that we replace n by $2n$. This is intuitive since all we did was double the sample size.

Therefore, to conclude $\bar{X}_{2n} \sim N\left(\mu, \frac{\sigma^2}{2n}\right)$

Part (d)

The 95% confidence interval for $\bar{X}_{2n}$ is: $\bar{X}_{2n} \pm \sqrt{\frac{\sigma^2 + \tau^2}{4n}}$. The underlying logic and methods don't change between parts (b) and (d). We can view the larger sample of size $2n$ as being all there is to talk about, and **not** being composed of two smaller subsamples. Its confidence interval has the same structure as in part (b): the estimate plus/minus the standard error. However, the estimate is $\bar{X}_{2n}$, the variance is the pooled or sum variances of each of the two smaller samples $\sigma^2 + \tau^2$, and the total size is $4n$.

Problem 2

In this experiment, we are asked to analyze the efficacy of only *one* television program based on its length in minutes. The “setting” of the experiment is the TV advertisement and the data that we analyze are the minutes that constitute the duration of the advertisement.

We are told that the treatment effect of the program, the amount of increased sales, is normally distributed in the following way:

$$X \sim N(\mu = \$500,000, \sigma^2 = \$200,000^2)$$

Since we are asked to find the expected net gain for purchasing 20 minutes of ads, we create a new random variable, S , which stands for total sales from 20 minutes of advertisement, and is defined as $S = 20 \cdot X$. That is, scaling up X by 20. This means:

$$S \sim N(\mu = \$10,000,000, \sigma^2 = \$16,000,000,000)$$

Expected gain, $E(G)$ is defined to be the expected revenue minus the expected cost. The expected revenue is simply $E(S)$ which is \$1,000,000. The expected cost can be found by taking the cost per minute and multiplying by the number of minutes, which for us is 20. Therefore, the expected cost is \$6,000,000. As such, the expected gain is:

$$E(G) = E(R) - E(C) = \$10,000,000 - \$6,000,000 = \$4,000,000$$

Problem 3

Part (a)

```
set.seed(12345)
included <- 0
n <- 100
for (i in 1:1000) {
  x <- runif(n, min = 0, max = 10)
  epsilon <- rnorm(n, mean = 0, sd = 1)
  y <- 2 + (3 * x) + epsilon

  regression <- lm(y ~ x)

  if (3 >= coef(regression)["x"]-2*summary(regression)$coefficients[2,2]
    & 3 <= coef(regression)["x"]+2*summary(regression)$coefficients[2,2])
  {
    included <- included + 1
  }
}
included
```

[1] 959

Upon running the above code, the number of times that $\beta_1 = 3$ occurred in the interval $b_1 \pm 2 \cdot \hat{se}(b_1)$ is 959. Therefore, the answer to part (a) is **TRUE**.

This makes mathematical sense as the confidence interval we test against is slightly wider than the 95% confidence interval. We should expect to see that, on average, 95% of the confidence intervals that we generate to contain the true parameter.

Part (b)

```
set.seed(12345)
included <- 0
n <- 100
for (i in 1:1000) {
  x <- runif(n, min = 0, max = 10)
  epsilon <- runif(n, min = -3, max = 3)
```

```

y <- 2 + (3 * x) + epsilon

regression <- lm(y ~ x)

if (3 >= coef(regression)["x"]-2*summary(regression)$coefficients[2,2]
& 3 <= coef(regression)["x"]+2*summary(regression)$coefficients[2,2])
{
  included <- included + 1
}
}
included

```

[1] 958

Changing the distribution of the error terms did not change the fact that 95% of the generated confidence intervals did contain the true population parameter of 3. Therefore, the answer to part (b) is TRUE.

Part (c)

```

set.seed(12345)
included <- 0
n <- 15
for (i in 1:1000) {
  x <- runif(n, min = 0, max = 10)
  epsilon <- rnorm(n, mean = 0, sd = 1)
  y <- 2 + (3 * x) + epsilon

  regression <- lm(y ~ x)

  if (3 >= coef(regression)["x"]-2*summary(regression)$coefficients[2,2]
& 3 <= coef(regression)["x"]+2*summary(regression)$coefficients[2,2])
  {
    included <- included + 1
  }
}
included

```

[1] 920

Things change when we lower the sample size per loop. Now, when we drop n to 15, we decrease the chance for our sample to contain the true population parameter. We have increased uncertainty about where the parameter lies, and therefore fewer intervals will contain said parameter. Therefore, the answer to part (c) is **FALSE**.

Problem 4

Parts (a), (b), (c), and (d)

```
female_adelie_penguins <- subset(penguins, species == "Adelie" & sex == "female")
regression <- lm(body_mass_g ~ bill_length_mm, data = female_adelie_penguins)
summary(regression)
```

Call:

```
lm(formula = body_mass_g ~ bill_length_mm, data = female_adelie_penguins)
```

Residuals:

Min	1Q	Median	3Q	Max
-501.73	-202.36	18.91	194.07	478.26

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2527.41	579.37	4.362	4.28e-05 ***
bill_length_mm	22.58	15.53	1.454	0.15

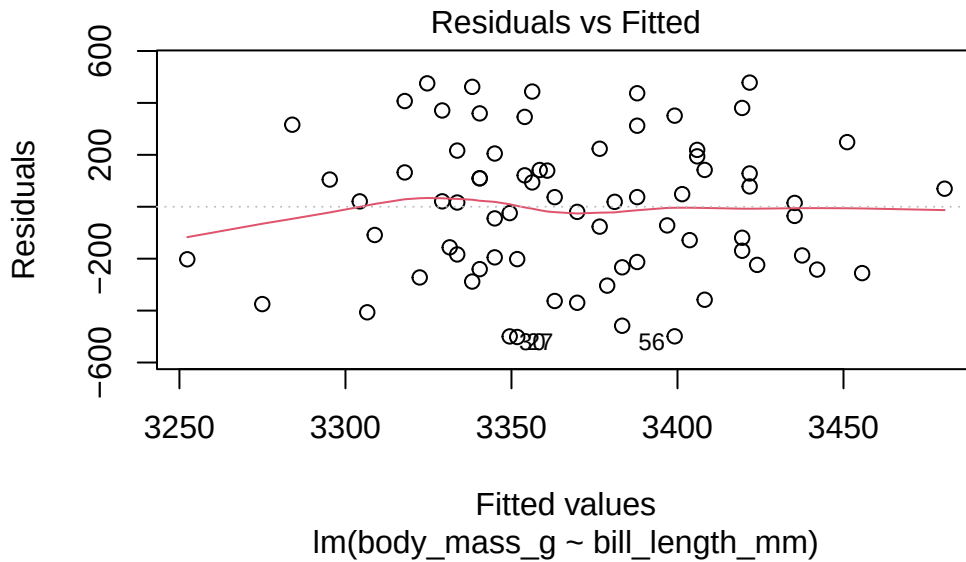
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 267.3 on 71 degrees of freedom

Multiple R-squared: 0.02893, Adjusted R-squared: 0.01526

F-statistic: 2.115 on 1 and 71 DF, p-value: 0.1502

```
plot(regression, which = 1)
```



```
predict(regression, data.frame(bill_length_mm = 40), interval = "confidence", 0.95)
```

```
$fit
      fit      lwr      upr
1 3430.771 3325.407 3536.135

$se.fit
[1] 52.84196

$df
[1] 71

$residual.scale
[1] 267.3175
```

```
predict(regression, data.frame(bill_length_mm = 40), interval = "prediction", 0.95)
```

```
$fit
      fit      lwr      upr
1 3430.771 2887.441 3974.101
```

```
$se.fit  
[1] 52.84196
```

```
$df  
[1] 71
```

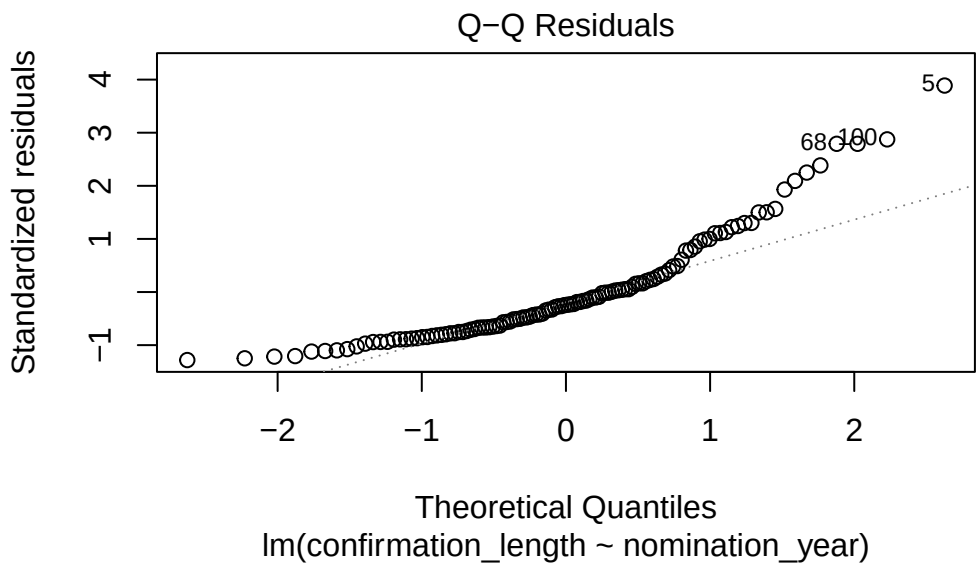
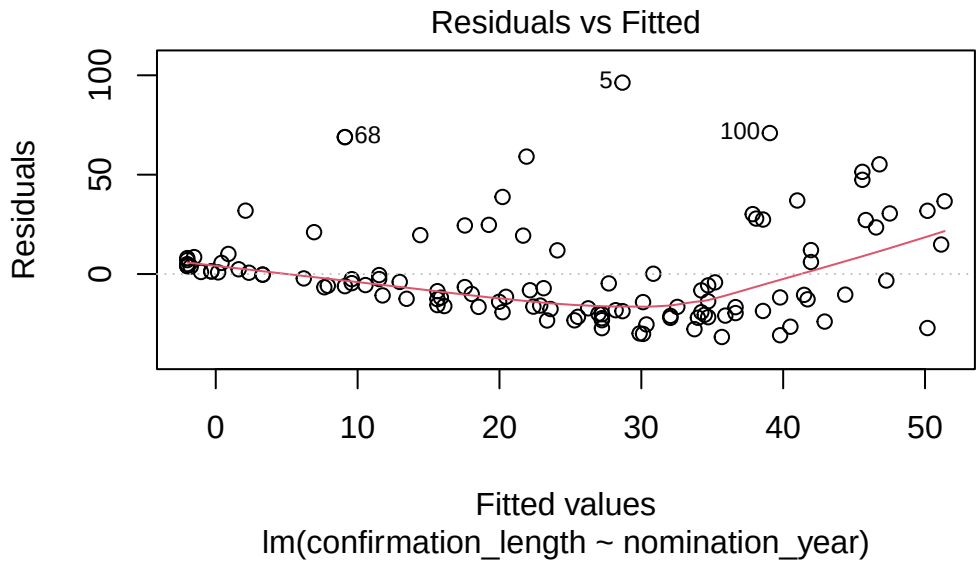
```
$residual.scale  
[1] 267.3175
```

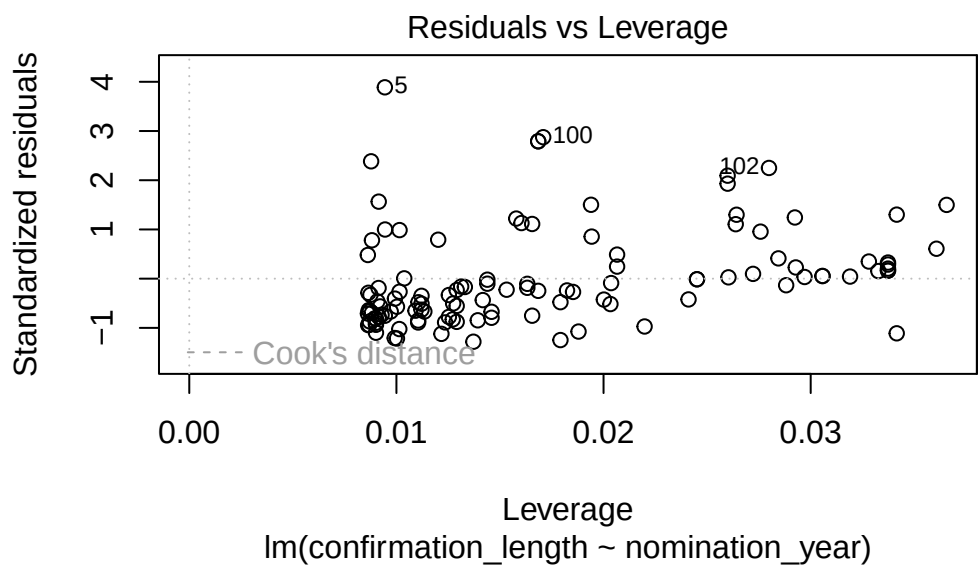
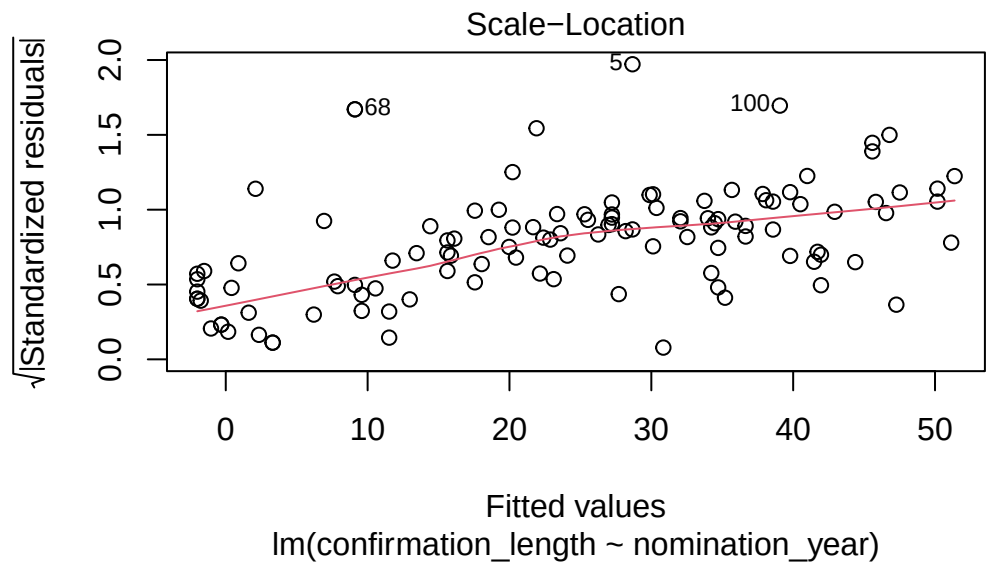
The data points appear to be evenly above and below the the zero-line as well as being distributed throughout the plot. This provides evidence that the data is well behaved for linear regression.

Problem 5

Part (a)

```
judge_appt_cleaned <- judges_appointments %>%  
  filter(court_type == "USSC") %>%  
  mutate(  
    nomination_date = mdy(nomination_date),  
    commission_date = mdy(commission_date),  
    confirmation_length = as.numeric(commission_date - nomination_date),  
    nomination_year = year(nomination_date)  
  ) %>%  
  filter(!is.na(confirmation_length))  
  
plot(regression <- lm(confirmation_length ~ nomination_year,  
  data = judge_appt_cleaned))
```





Parts (b) and (c)

```
summary(regression <- lm(confirmation_length ~ nomination_year, data = judge_appt_cleaned))
```

Call:

```
lm(formula = confirmation_length ~ nomination_year, data = judge_appt_cleaned)
```

Residuals:

Min	1Q	Median	3Q	Max
-31.678	-17.747	-6.001	8.137	96.328

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-434.19232	69.60304	-6.238	7.75e-09 ***
nomination_year	0.24158	0.03668	6.586	1.45e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24.89 on 114 degrees of freedom

Multiple R-squared: 0.2756, Adjusted R-squared: 0.2693

F-statistic: 43.38 on 1 and 114 DF, p-value: 1.448e-09

```
confint(regression, param = "nomination_year", level = 0.95)
```

	2.5 %	97.5 %
(Intercept)	-572.0754200	-296.3092295
nomination_year	0.1689168	0.3142406

The estimated slope parameter for this linear regression is 0.24158 days of confirmation per year. I interpret the slope to be that the confirmation length gets extended by about 0.24158 days, on average, every year.

For this confidence interval to be valid, we must assume: 1. linearity between the treatment and response variables which is demonstrated via the residuals-vs-fitted plot. 2. Independence, which is tricky to accurately confirm, since any given political climate is probably going to have an influence on the confirmation process as well as the changes from SCOTUS Justice to SCOTUS Justice based on the political party in power. 3. Normality in the residuals which can be checked from a Quantile-Quantile plot 4. Uniform variances across time. This can be analyzed via a scale-location plot.

Some of these plots suggest that these assumptions do not appear to hold for this data set. For example, the residuals-vs-fitted plot has some curvature to it and the Q-Q plot also shows deviations from the diagonal.

Part (d)

I think the biggest hurdle that this linear analysis faces is time. Specifically, how the political climate has changed throughout the entire history of the United States of America. Modern commentators suggest that the US is living through an extremely strained and polarized time, which probably has an outsized negative effect on confirmation length.